Answer the following questions:

1. **What is a search tree in data structures?**

- A search tree refers to a type of data structure that makes use of a tree for the purpose of effectively arranging data in a hierarchical manner so that retrieval, insertion and deletion of the data can easily be achieved. A typical representation of a search tree can be seen in a binary search tree where each of its nodes perceives left-related child nodes as having less value than itself while right-related child nodes are said to have more value than itself.

2. **What are the different types of search algorithms in data structures? Differentiate each type of search.**

- There are two main categories into which search algorithms fall: (1) Sequential or Linear Search and (2) Interval or Binary Search. In a sequential search, it inspect every item linearly starting from the first item to the last item. This means Sequential Search can be applied to both ordered and unordered datasets. Furthermore, the time required to execute Sequential Search is O(n). On the other hand, the Interval or Binary Search problem is one where the problem is solved in a divide-and-conquer style. Thus, it is possible to apply Binary Search only on ordered lists, and the computational complexity for this search is O(log n).

3. **What operations/implementations can be performed using binary and linear search operations?**

- Linear search algorithms are applicable on unordered types of structures, such as linked lists or raw streams. This means that the linear search is useful in cases of searching unindexed data, finding the minimum and maximum value, and counting the instances. On the other hand, binary search requires a specific kind of data, namely sorted data that provides direct access to information.

4. **What are the advantages in using a binary search tree as a data structure?**

- A Binary Search Tree has an average O(log n) time complexity for search, insertion, and deletion operations and dynamically allocates memory instead of having size limits. Unlike arrays, inserting or deleting nodes does not require shifting elements, and an in-order traversal yields all items in sorted order in O(n) time.

5. **Give an example program using binary search and Linear search.**

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